Telemetry Peripheral Connector

SM06B-GHS-TB J1

BAS516 D1

BZX585-B5V6.115 D2

NL03KT330 L1

C1 10uF

JP1 RST_EXT

PWR_FLAG

R1 100R RX0

R4 100R TX0

RTS

CTS

MP

GND

+VSW

1uH – DFE252012P-1R0M=P2 L2

C5 22u

TPS631000DRLR U1

VIN LX1 LX2 VOUT FB GND MODE EN

+3.3V

R10 511k

R11 91k

3v3l_FB

C6 47u

C7 10u

D5 RED

R14 180R

1uH – DFE252012P-1R0M=P2 L3

C13 22u

+5V

R22 806k

R23 91k

5v_FB

C14 47u

C15 10u

D7 RED

R24 180R

U3

THUNDERFLY G1

RST_EXT

C2 4u7

RST_PRG

R5 10k

Y1 8 MHz

C3 15p

C4 15p

C8 10u

C9 100nF

PWR_FLAG

U2 ATMEGA1284P-AU

RESET XTAL1 XTAL2 AREF VCC AVCC GND

PA0 A0 PA1 A1 PA2 PA3 EXT_DETECTION PA4 CTS PA5 RTS PA6 BTN_USER_A PA7 BTN_USER_B

PB0 B0 PB1 B1 PB2 PB3 CS2 PB4 CS1 PB5 MOSI PB6 MISO PB7 SCLK

PC0 SCL PC1 SDA PC2 C0 PC3 PC4 PC5 LED1 PC6 LED2 PC7 LED3

PD0 RX0 PD1 TX0 PD2 RX1 PD3 TX1 PD4 TIMEPULSE PD5 EXTINT PD6 GEO_STAT PD7 C1

C10 100nF C11 100nF C12 100nF

TP1 TP2

R12 10k R13 10k

J3 MISO SCLK RST_PRG MOSI

SW1 SKRTLAE010 R2 10k BTN_USER_A

SW2 SKRTLAE010 R3 10k BTN_USER_B

M1 M2 M3 M4 GND MOUNTING HOLES

FID1 Fiducial FID2 Fiducial FID3 Fiducial

Transistors may be removed in future revision,

D3 BLUE R6 51R R7 10k Q1 BC817-25 LED2

D6 GREEN R15 18R LED3

D4 RED R8 82R R9 10k Q2 BC817-25 LED1

J2 SMOBB-GHS-TB ThunderFly TFGPS Payload Connector

TIMEPULSE EXTINT GEO_STAT SDA SCL TX1 RX1 R17 R18 R19 R20 R21 100R MP

J4 SM04B-GHS-TB I²C – Dronecode Autopilot Connector Standard

J6 SM07B-GHS-TB SPI – Dronecode Autopilot Connector Standard

SCLK MISO MOSI CS1 CS2 R25 R26 R27 R28 R29 R30 100R

J5 HEADER_2x15

A0 A1 B1 B0 C1 C0 RX1 TX1 MOSI MISO SCLK RST_EXT EXT_DETECTION R26 10k

TFUNIPAYLOAD01B

Universal interface for TF-ATMON sensor payload.

Author: ThunderFly s.r.o.

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THUNDERFLY